

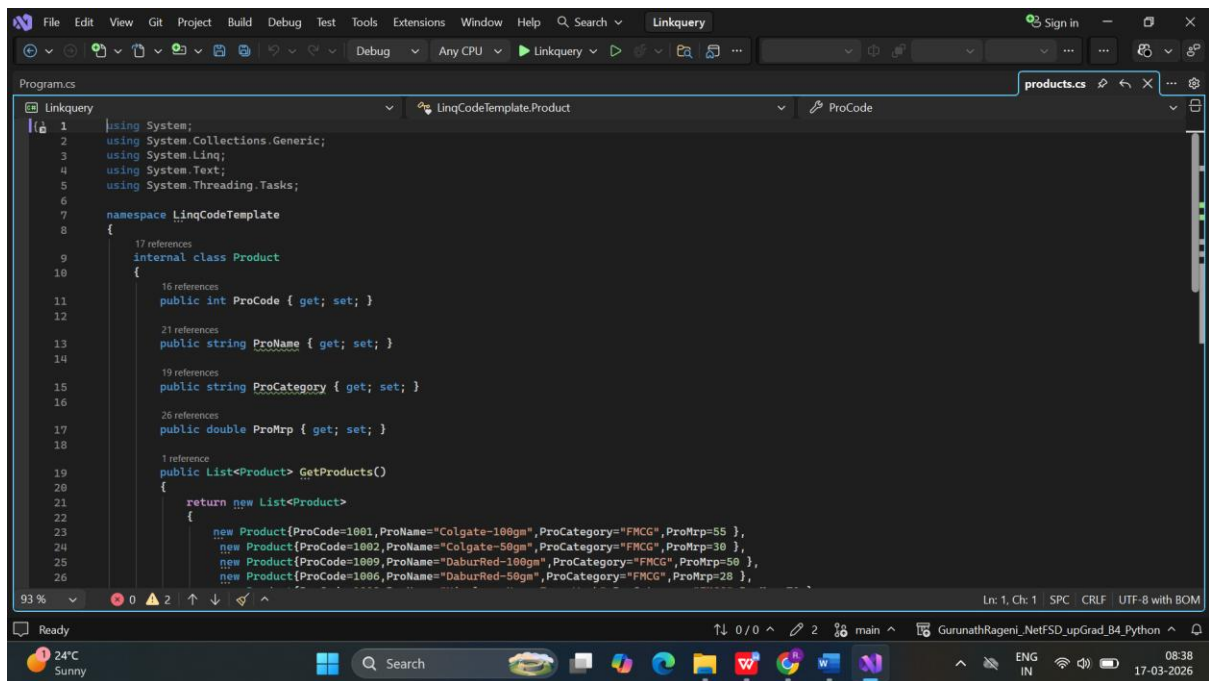
Please download and refer shared Code template ([LinqCodeTemplate](#)) and solve problems as given below.

(Please Refer Problem-1 Solved in the Code Template and solve rest of other problems in the same project accordingly)

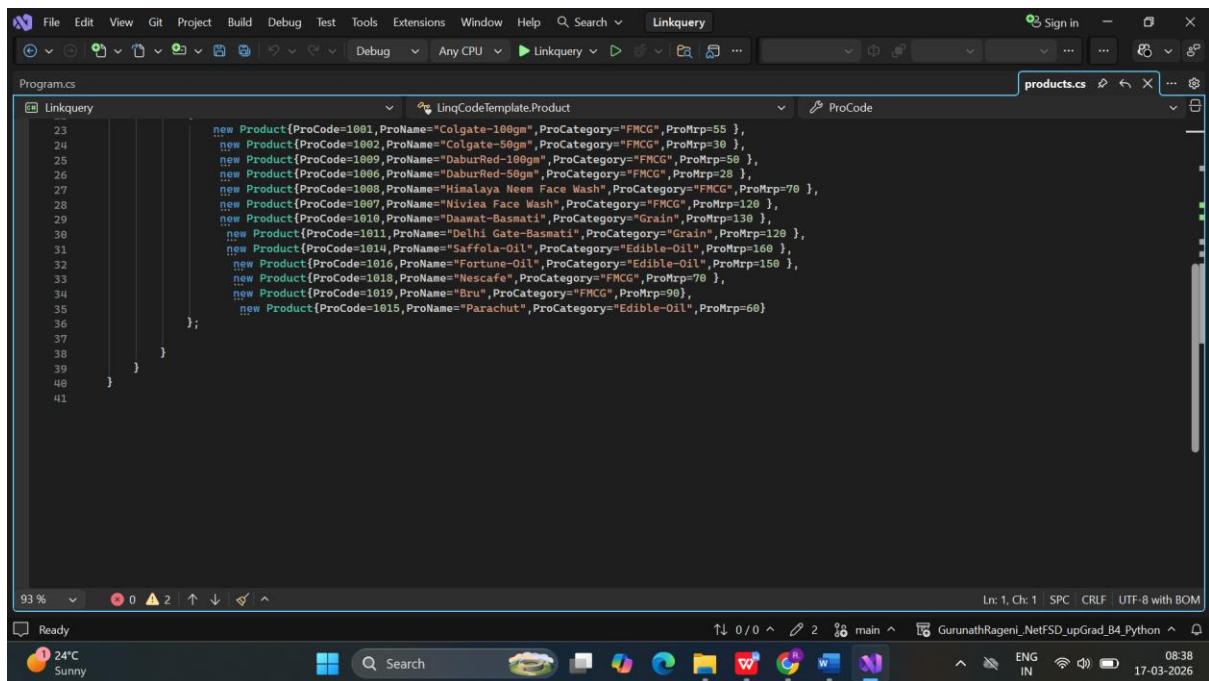
Problem Level- 1 and 2:

1. Write a LINQ query to search and display all products with category “FMCG”.
2. Write a LINQ query to search and display all products with category “Grain”.
3. Write a LINQ query to sort products in ascending order by product code.
4. Write a LINQ query to sort products in ascending order by product Category.
5. Write a LINQ query to sort products in ascending order by product Mrp.
6. Write a LINQ query to sort products in descending order by product Mrp.
7. Write a LINQ query to display products group by product Category.
8. Write a LINQ query to display products group by product Mrp.
9. Write a LINQ query to display product detail with highest price in FMCG category.
10. Write a LINQ query to display count of total products.
11. Write a LINQ query to display count of total products with category FMCG.
12. Write a LINQ query to display Max price.
13. Write a LINQ query to display Min price.
14. Write a LINQ query to display whether all products are below Mrp Rs.30 or not.
15. Write a LINQ query to display whether any products are below Mrp Rs.30 or not.

Code Screenshot:



```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6
7 namespace LinkCodeTemplate
8 {
9     17 references
10     internal class Product
11     {
12         16 references
13         public int ProCode { get; set; }
14
15         21 references
16         public string ProName { get; set; }
17
18         19 references
19         public string ProCategory { get; set; }
20
21         26 references
22         public double ProMrp { get; set; }
23
24         1 reference
25         public List<Product> GetProducts()
26         {
27             return new List<Product>
28             {
29                 new Product{ProCode=1001,ProName="Colgate-100gm",ProCategory="FMCG",ProMrp=55 },
30                 new Product{ProCode=1002,ProName="Colgate-50gm",ProCategory="FMCG",ProMrp=30 },
31                 new Product{ProCode=1009,ProName="DaburRed-100gm",ProCategory="FMCG",ProMrp=50 },
32                 new Product{ProCode=1006,ProName="DaburRed-50gm",ProCategory="FMCG",ProMrp=28 },
33             }
34         }
35     }
36 }
```



```
23 new Product{ProCode=1001,ProName="Colgate-100gm",ProCategory="FMCG",ProMrp=55 },
24 new Product{ProCode=1002,ProName="Colgate-50gm",ProCategory="FMCG",ProMrp=30 },
25 new Product{ProCode=1009,ProName="DaburRed-100gm",ProCategory="FMCG",ProMrp=50 },
26 new Product{ProCode=1006,ProName="DaburRed-50gm",ProCategory="FMCG",ProMrp=28 },
27 new Product{ProCode=1008,ProName="Himalaya Neem Face Wash",ProCategory="FMCG",ProMrp=70 },
28 new Product{ProCode=1007,ProName="Nivea Face Wash",ProCategory="FMCG",ProMrp=120 },
29 new Product{ProCode=1010,ProName="Daawat-Basmati",ProCategory="Grain",ProMrp=130 },
30 new Product{ProCode=1011,ProName="Delhi Gate-Basmati",ProCategory="Grain",ProMrp=120 },
31 new Product{ProCode=1014,ProName="Saffola-Oil",ProCategory="Edible-Oil",ProMrp=160 },
32 new Product{ProCode=1016,ProName="Fortune-Oil",ProCategory="Edible-Oil",ProMrp=150 },
33 new Product{ProCode=1018,ProName="Nescafe",ProCategory="FMCG",ProMrp=70 },
34 new Product{ProCode=1019,ProName="Bru",ProCategory="FMCG",ProMrp=90 },
35 new Product{ProCode=1015,ProName="Parachut",ProCategory="Edible-Oil",ProMrp=60 }
36 }
37 }
38 }
39 }
40 }
41 }
```

```
File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Linkquery Sign in
Debug Any CPU Linkquery
Program.cs products.cs
Linkquery Program <top-level-statements-entry-point>
1 // 2. Products with category "Grain"
2 using LinkCodeTemplate;
3
4 Product product = new Product();
5 var products = product.GetProducts();
6 grainProducts = products.Where(p => p.ProCategory == "Grain");
7 Console.WriteLine("\nGrain Products:");
8 foreach (var item in grainProducts)
9 {
10     Console.WriteLine($"{item.ProCode}\t{item.ProName}\t{item.ProMrp}");
11 }
12
13 // 3. Sort by Product Code (Ascending)
14 var sortByCode = products.OrderBy(p => p.ProCode);
15 Console.WriteLine("\nSorted by Product Code:");
16 foreach (var item in sortByCode)
17 {
18     Console.WriteLine($"{item.ProCode}\t{item.ProName}");
19 }
20
21 // 4. Sort by Category (Ascending)
22 var sortByCategory = products.OrderBy(p => p.ProCategory);
23 Console.WriteLine("\nSorted by Category:");
24 foreach (var item in sortByCategory)
25 {
26     Console.WriteLine($"{item.ProCategory}\t{item.ProName}");
27 }
28
29 // 5. Sort by MRP (Ascending)
30
93 % 0 1 ↑ ↓ ↶ ↷ Ln: 5, Ch: 38 SPC CRLF UTF-8 with BOM
Ready 0/0 2 main GurunathRageni_NetFSD_upGrad_B4_Python 24°C Sunny 08:38 17-03-2026
```

```
File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Linkquery Sign in
Debug Any CPU Linkquery
Program.cs products.cs
Linkquery Program <top-level-statements-entry-point>
28 Console.WriteLine($"{item.ProCategory}\t{item.ProName}");
29 }
30
31 // 5. Sort by MRP (Ascending)
32 var sortByMrpAsc = products.OrderBy(p => p.ProMrp);
33 Console.WriteLine("\nSorted by MRP (Ascending):");
34 foreach (var item in sortByMrpAsc)
35 {
36     Console.WriteLine($"{item.ProName}\t{item.ProMrp}");
37 }
38
39 // 6. Sort by MRP (Descending)
40 var sortByMrpDesc = products.OrderByDescending(p => p.ProMrp);
41 Console.WriteLine("\nSorted by MRP (Descending):");
42 foreach (var item in sortByMrpDesc)
43 {
44     Console.WriteLine($"{item.ProName}\t{item.ProMrp}");
45 }
46
47 // 7. Group by Category
48 var groupByCategory = products.GroupBy(p => p.ProCategory);
49 Console.WriteLine("\nGrouped by Category:");
50 foreach (var group in groupByCategory)
51 {
52     Console.WriteLine($"{group.Key}");
53     foreach (var item in group)
54     {
55         Console.WriteLine($"{item.ProName}\t{item.ProMrp}");
56     }
57 }
58
93 % 0 1 ↑ ↓ ↶ ↷ Ln: 5, Ch: 38 SPC CRLF UTF-8 with BOM
Ready 0/0 2 main GurunathRageni_NetFSD_upGrad_B4_Python 24°C Sunny 08:38 17-03-2026
```

```
File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Linkquery Sign in
Debug Any CPU Linkquery products.cs
Program.cs
55 Console.WriteLine($"Category: {group.Key}");
56 foreach (var item in group)
57 {
58     Console.WriteLine($"{item.ProName}\t{item.ProMrp}");
59 }
60
61
62
63 // 8. Group by MRP
64 var groupByMrp = products.GroupBy(p => p.ProMrp);
65 Console.WriteLine($"Grouped by MRP:");
66 foreach (var group in groupByMrp)
67 {
68     Console.WriteLine($"MRP: {group.Key}");
69     foreach (var item in group)
70     {
71         Console.WriteLine($"{item.ProName}");
72     }
73 }
74
75
76 // 9. Highest price product in FMCG
77 var maxFmcg = products
78     .Where(p => p.ProCategory == "FMCG")
79     .OrderByDescending(p => p.ProMrp)
80     .FirstOrDefault();
81
82 Console.WriteLine($"Highest Price FMCG Product:");
83 Console.WriteLine($"{maxFmcg.ProName}\t{maxFmcg.ProMrp}");
84
85
86 // 10. Total product count
93 % Ln: 5, Ch: 38 SPC CRLF UTF-8 with BOM
Ready 0/0 2 main GurunathRageni_NetFSD_upGrad_B4_Python 24°C Sunny 08:38 17-03-2026
```

```
File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Linkquery Sign in
Debug Any CPU Linkquery products.cs
Program.cs
85
86 // 10. Total product count
87 var totalCount = products.Count();
88 Console.WriteLine($"Total Products: {totalCount}");
89
90
91 // 11. Count of FMCG products
92 var fmcgCount = products.Count(p => p.ProCategory == "FMCG");
93 Console.WriteLine($"FMCG Product Count: {fmcgCount}");
94
95
96 // 12. Max price
97 var maxPrice = products.Max(p => p.ProMrp);
98 Console.WriteLine($"Max Price: {maxPrice}");
99
100
101 // 13. Min price
102 var minPrice = products.Min(p => p.ProMrp);
103 Console.WriteLine($"Min Price: {minPrice}");
104
105
106 // 14. All products below Rs.30?
107 var allBelow30 = products.All(p => p.ProMrp < 30);
108 Console.WriteLine($"All products below Rs.30: {allBelow30}");
109
110
111 // 15. Any product below Rs.30?
112 var anyBelow30 = products.Any(p => p.ProMrp < 30);
113 Console.WriteLine($"Any product below Rs.30: {anyBelow30}");
93 % Ln: 5, Ch: 38 SPC CRLF UTF-8 with BOM
Ready 0/0 2 main GurunathRageni_NetFSD_upGrad_B4_Python 24°C Sunny 08:38 17-03-2026
```

Code output:

```
Microsoft Visual Studio Debu: x + v
Grain Products:
1010 Daawat-Basmati 130
1011 Delhi Gate-Basmati 120

Sorted by Product Code:
1001 Colgate-100gm
1002 Colgate-50gm
1006 DaburRed-50gm
1007 Nivia Face Wash
1008 Himalaya Neem Face Wash
1009 DaburRed-100gm
1010 Daawat-Basmati
1011 Delhi Gate-Basmati
1014 Saffola-Oil
1015 Parachut
1016 Fortune-Oil
1018 Nescafe
1019 Bru

Sorted by Category:
Edible-Oil Saffola-Oil
Edible-Oil Fortune-Oil
Edible-Oil Parachut
FMCG Colgate-100gm
FMCG Colgate-50gm
FMCG DaburRed-100gm
FMCG DaburRed-50gm
FMCG Himalaya Neem Face Wash
FMCG Nivia Face Wash
FMCG Nescafe
FMCG Bru
Grain Daawat-Basmati

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```

```
Microsoft Visual Studio Debu: x + v
Sorted by MRP (Descending):
Saffola-Oil 160
Fortune-Oil 150
Daawat-Basmati 130
Nivia Face Wash 120
Delhi Gate-Basmati 120
Bru 90
Himalaya Neem Face Wash 70
Nescafe 70
Parachut 60
Colgate-100gm 55
DaburRed-100gm 50
Colgate-50gm 30
DaburRed-50gm 28

Grouped by Category:

Category: FMCG
Colgate-100gm 55
Colgate-50gm 30
DaburRed-100gm 50
DaburRed-50gm 28
Himalaya Neem Face Wash 70
Nivia Face Wash 120
Nescafe 70
Bru 90

Category: Grain
Daawat-Basmati 130
Delhi Gate-Basmati 120

Category: Edible-Oil

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```

```
Microsoft Visual Studio Debug Console
+ -

Grouped by MRP:

MRP: 55
Colgate-100gm

MRP: 30
Colgate-50gm

MRP: 50
DaburRed-100gm

MRP: 28
DaburRed-50gm

MRP: 70
Himalaya Neem Face Wash
Nescafe

MRP: 120
Nivea Face Wash
Delhi Gate-Basmati

MRP: 130
Daawat-Basmati

MRP: 160
Saffola-Oil

MRP: 150
Fortune-Oil

MRP: 90
Bru

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```

```
Microsoft Visual Studio Debug Console
+ -

MRP: 130
Daawat-Basmati

MRP: 160
Saffola-Oil

MRP: 150
Fortune-Oil

MRP: 90
Bru

MRP: 60
Parachut

Highest Price FMCG Product:
Nivea Face Wash      120

Total Products: 13
FMCG Product Count: 8

Max Price: 160
Min Price: 28

All products below Rs.30: False
Any product below Rs.30: True

C:\Users\gurun\OneDrive\Desktop\GurunathRageni_.NetFSD_upGrad_B4_Python\Week_5\Day_1_Handson_Csharp\Linquery\bin\Debug\net10.0\Linkq
uery.exe (process 22328) exited with code 0 (0x0).
Press any key to close this window . . .

24°C Sunny 08:41 17-03-2026
```